

Schematic: reComputer J401

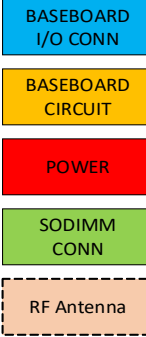
SHEET	SHEET NAME
01	Table of Contents
02	Block Diagram
03	Power Tree
04	SODIMM Connector 1/3
05	SODIMM Connector 2/3
06	SODIMM Connector 3/3
07	Power In & 5V & 3.3V_MCU
08	3.3V & 1.8V & 1.05V
09	Button MCU For Power Up
10	USB3.2 HUB
11	USB3.2 Type-A x2 (A)
12	USB3.2 Type-A x2 (B)
13	HDMI
14	CAM Connector
15	Gigabit Ethernet, Type C
16	M.2 KEY-E (WIFI/BT)
17	M.2 KEY-M (NVME)
18	40Pin GPIO Expansion Header
19	Fan, Debug and EEPROM

Revision History

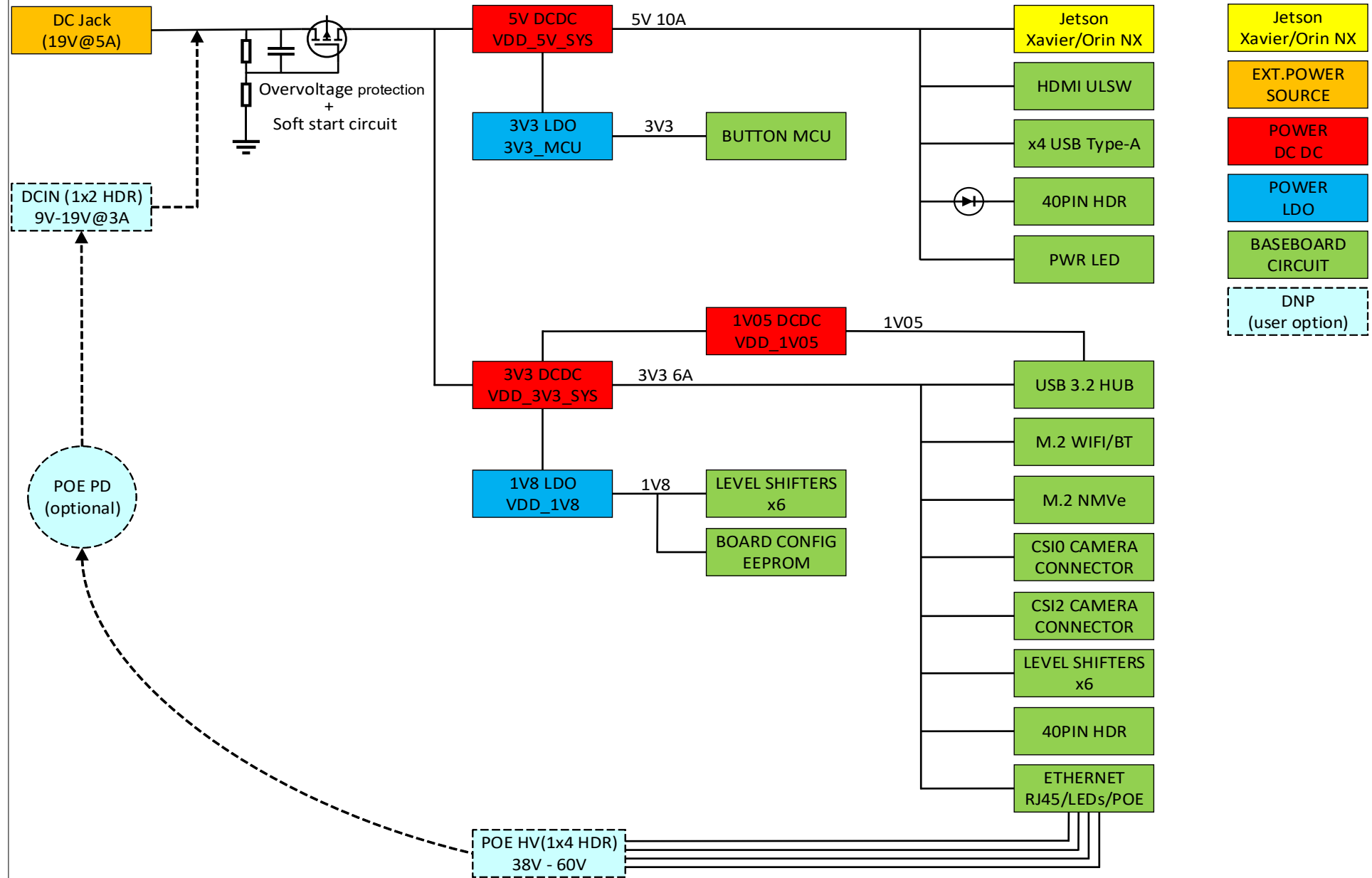
DATE	REVISION	DESCRIPTION
01/05/2023	reComputer J401_V1.0_SCH_230105	Initial Release.
08/09/2024	reComputer J401_V1.0_SCH_240802	Page 09:Changed X1 from 5YAA12000182TFA0Q3 to M12000I382. Changed C147 and C157 from 27pF to 33pF.
12/13/2024	reComputer J401B	Page 11:Add J21 USB 3.0 connector for 4G hat. DNP J4.

Block Diagram

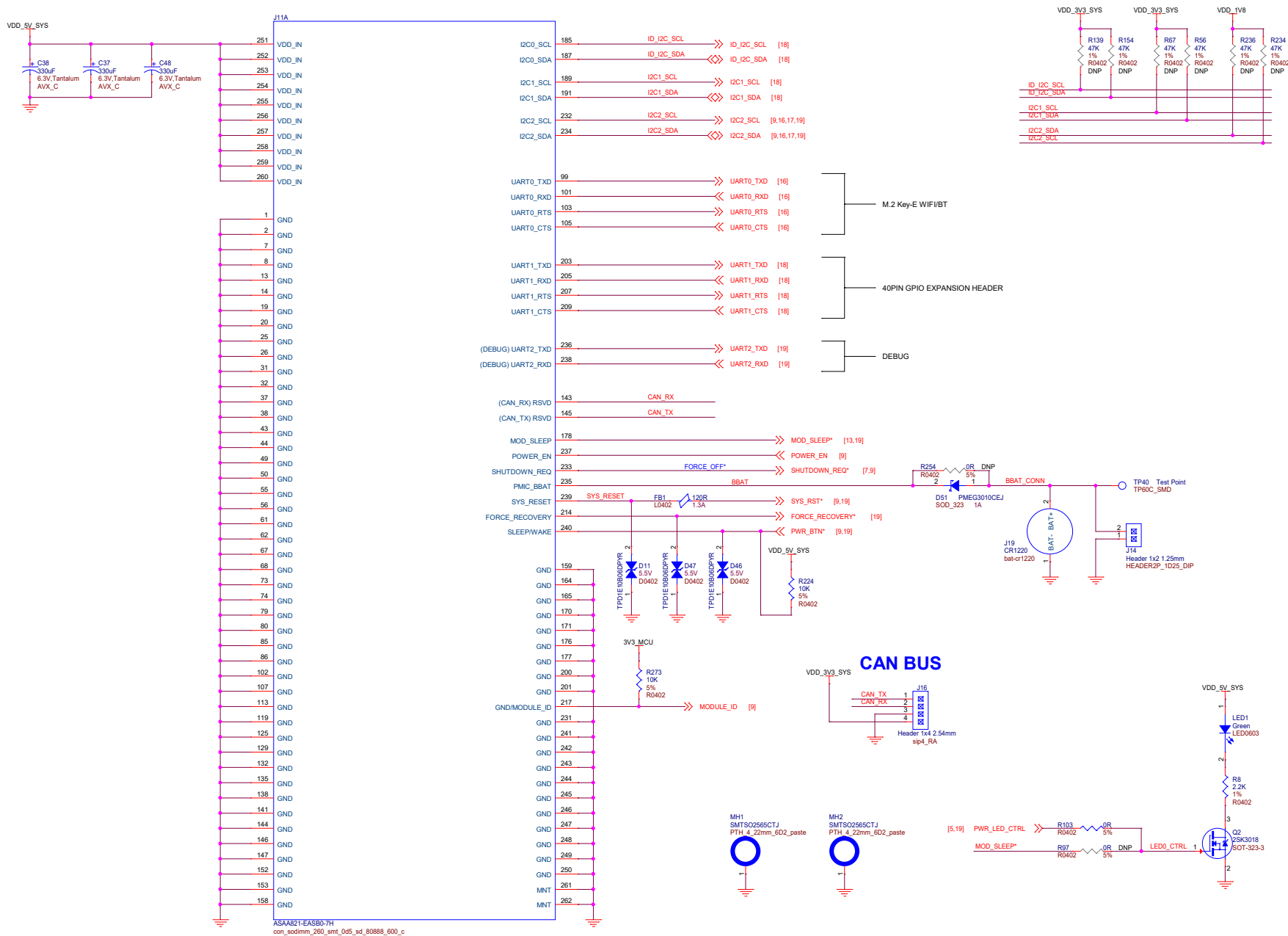
Map Legend:



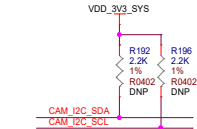
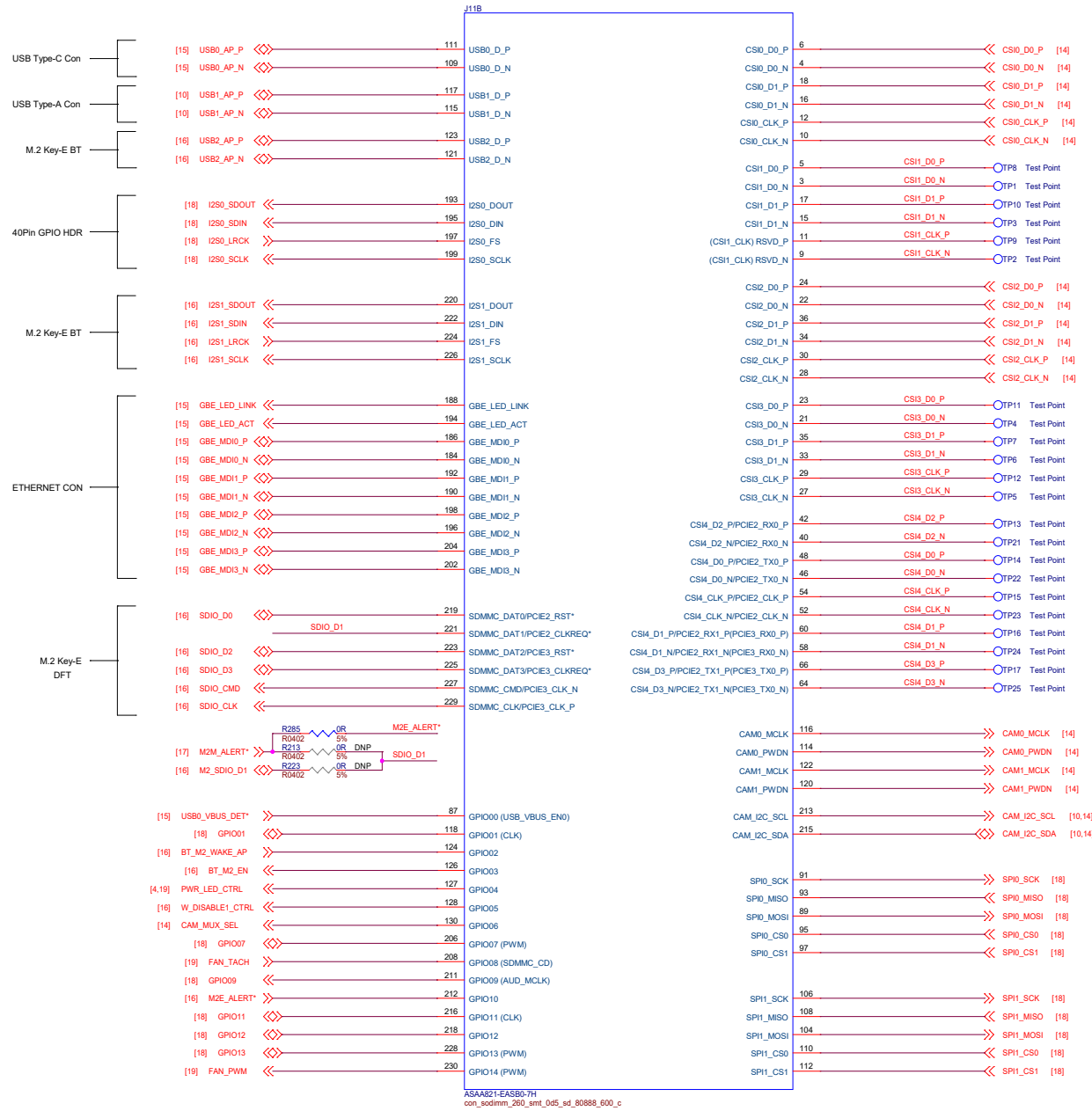
Power Topology



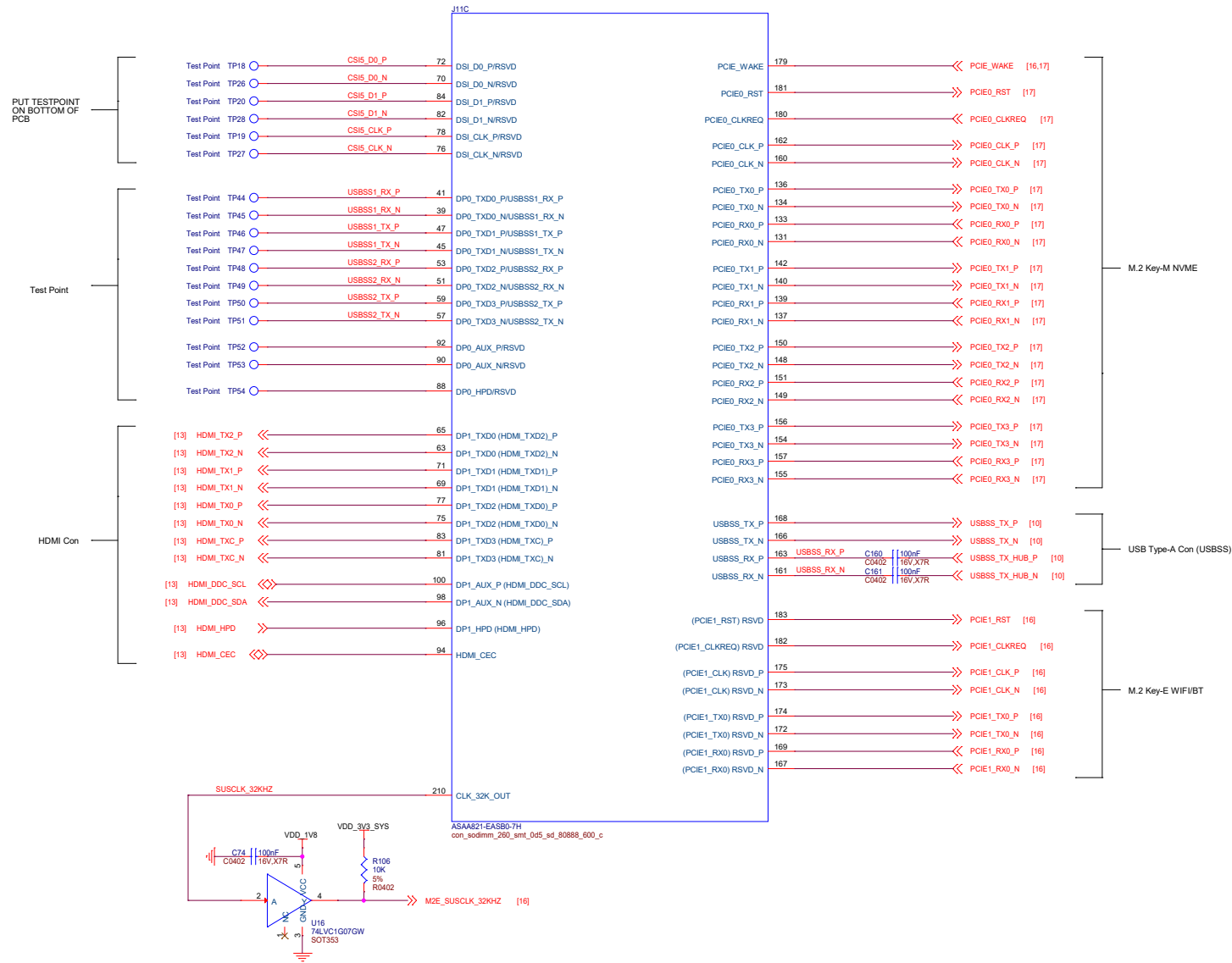
UART, I2C, CAN and GPIOs

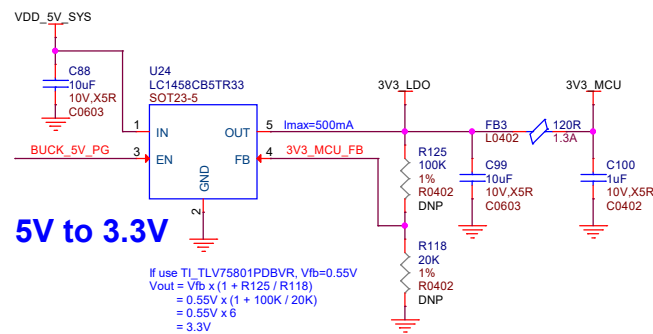
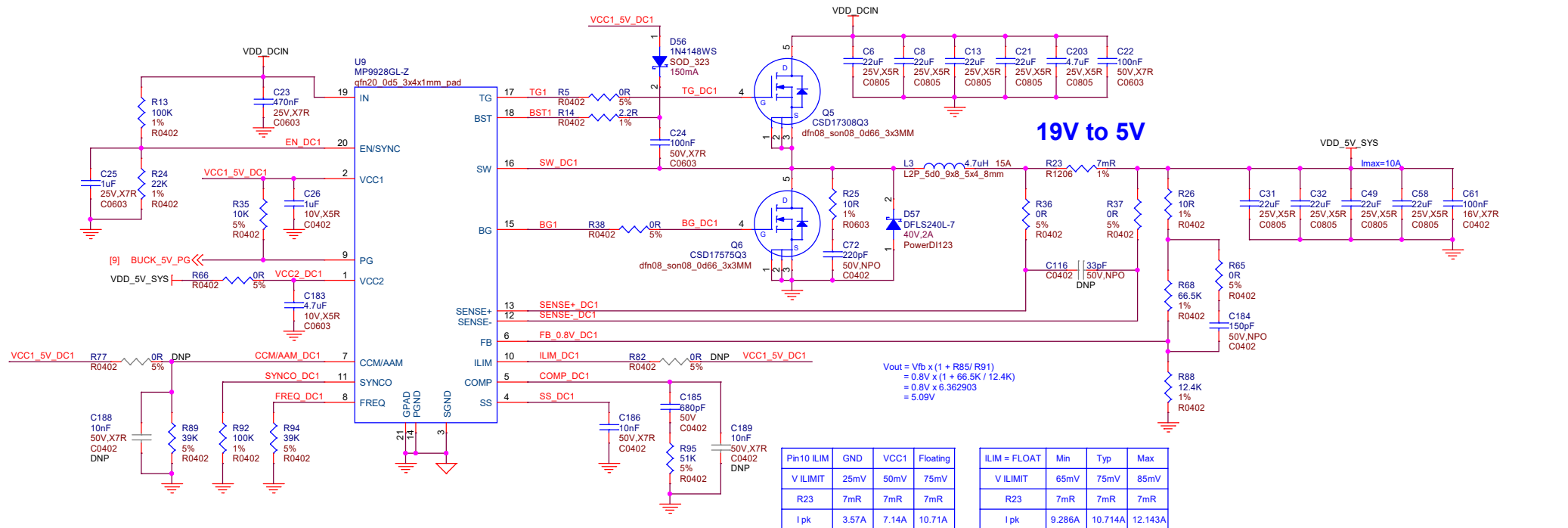
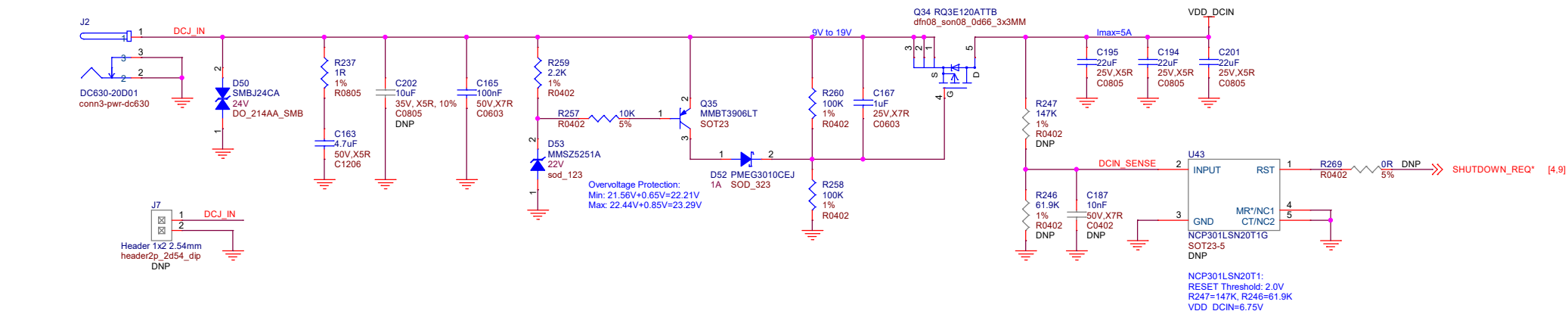


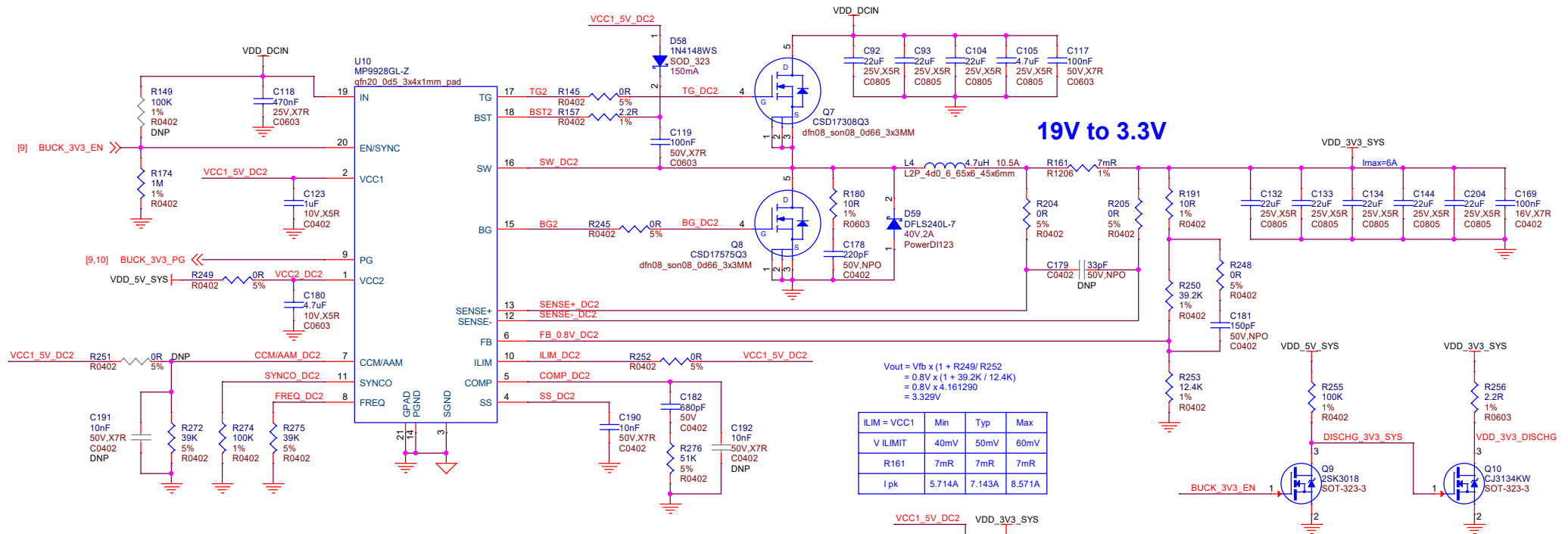
GPIOs, USB2.0, I2S, SPI, GBE and CSI



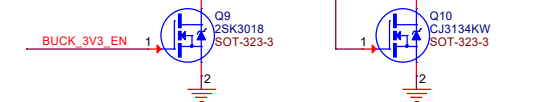
HDMI, DP, DSI, USBSS and PCIe



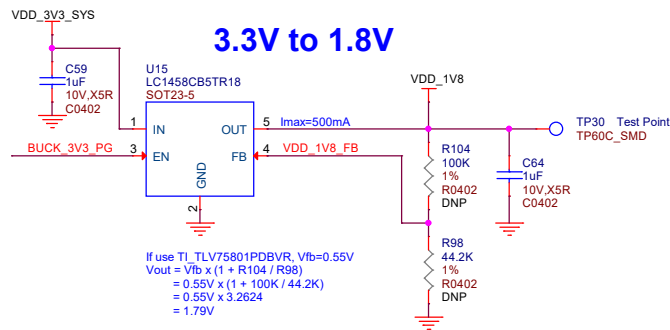




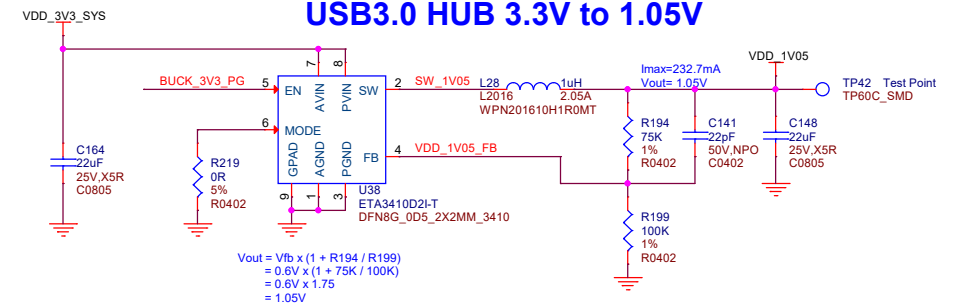
VDD_3V3_SYS RAIL DISCHARGE



3.3V to 1.8V

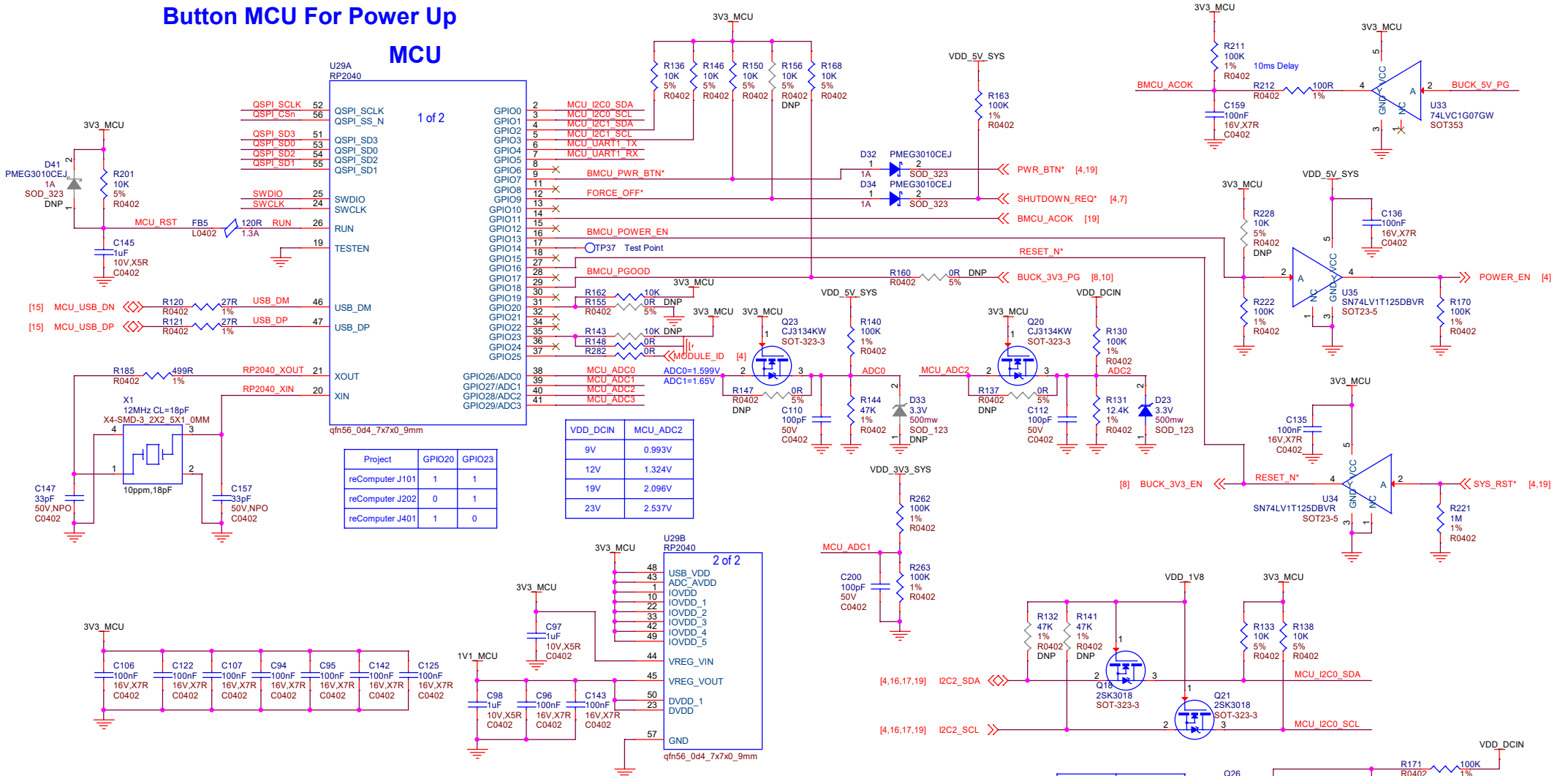


USB3.0 HUB 3.3V to 1.05V

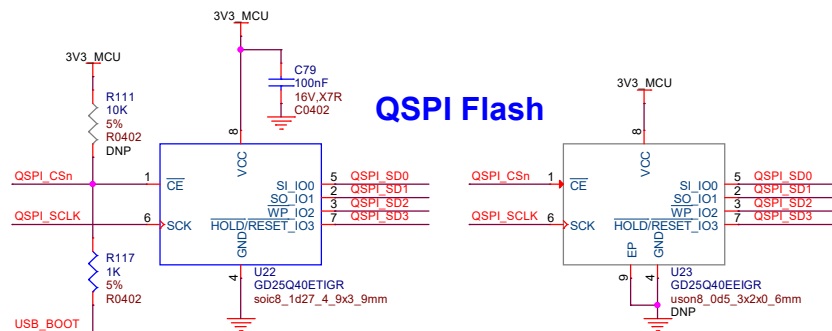


Button MCU For Power Up

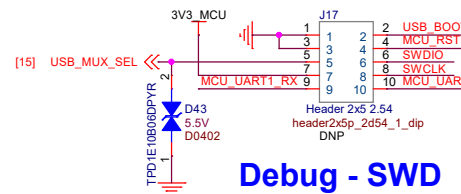
MCU



QSPI Flash



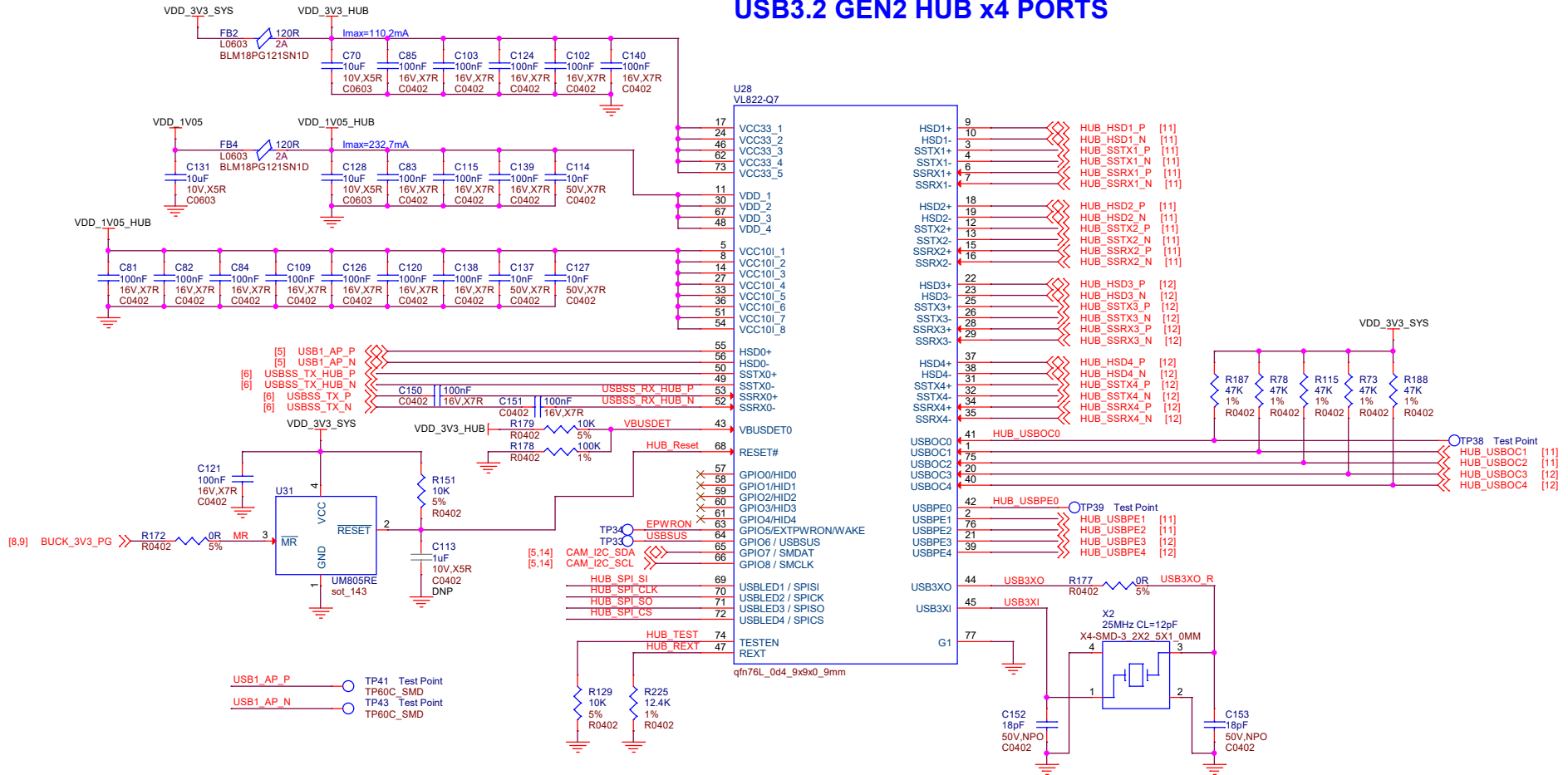
USB BOOT



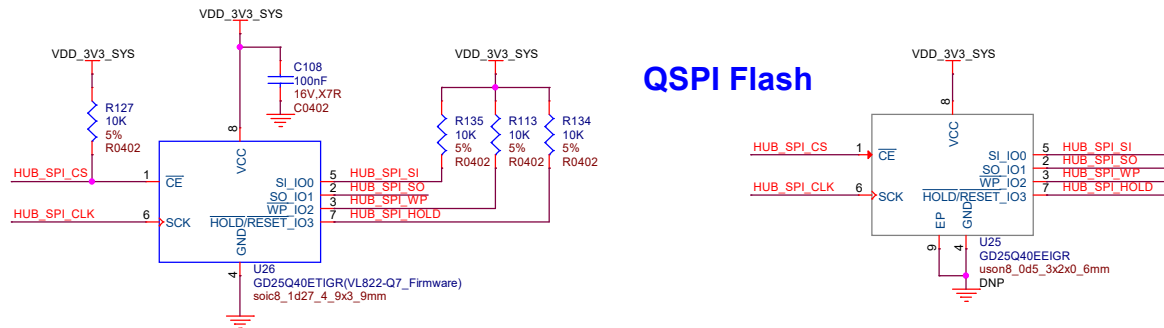
Debug - SWD

VDD_DCIN	FORCE_OFF*
9V	1.725V
12V	2.029V
19V	2.354V
23V	XXXXX

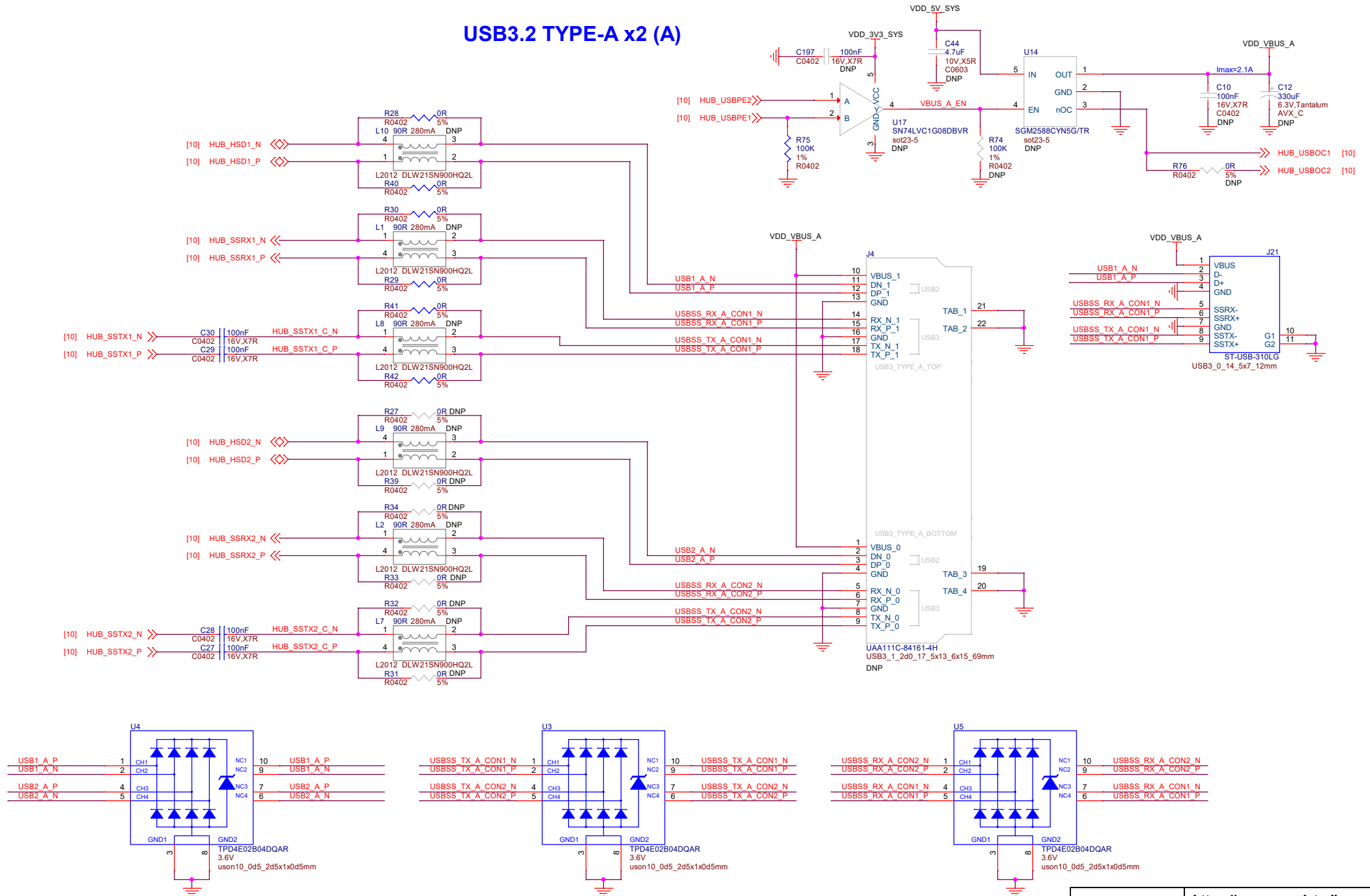
USB3.2 GEN2 HUB x4 PORTS



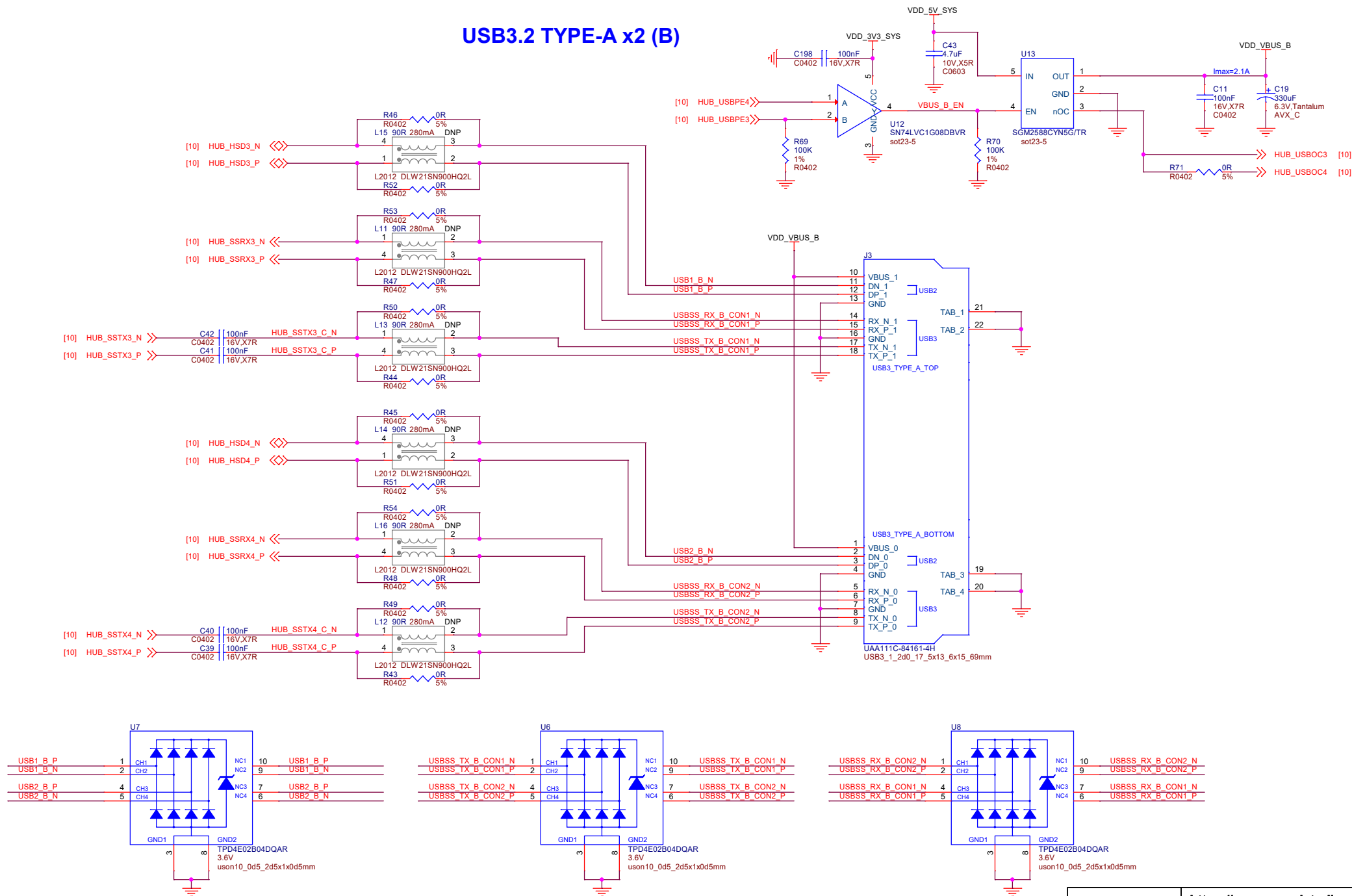
QSPI Flash



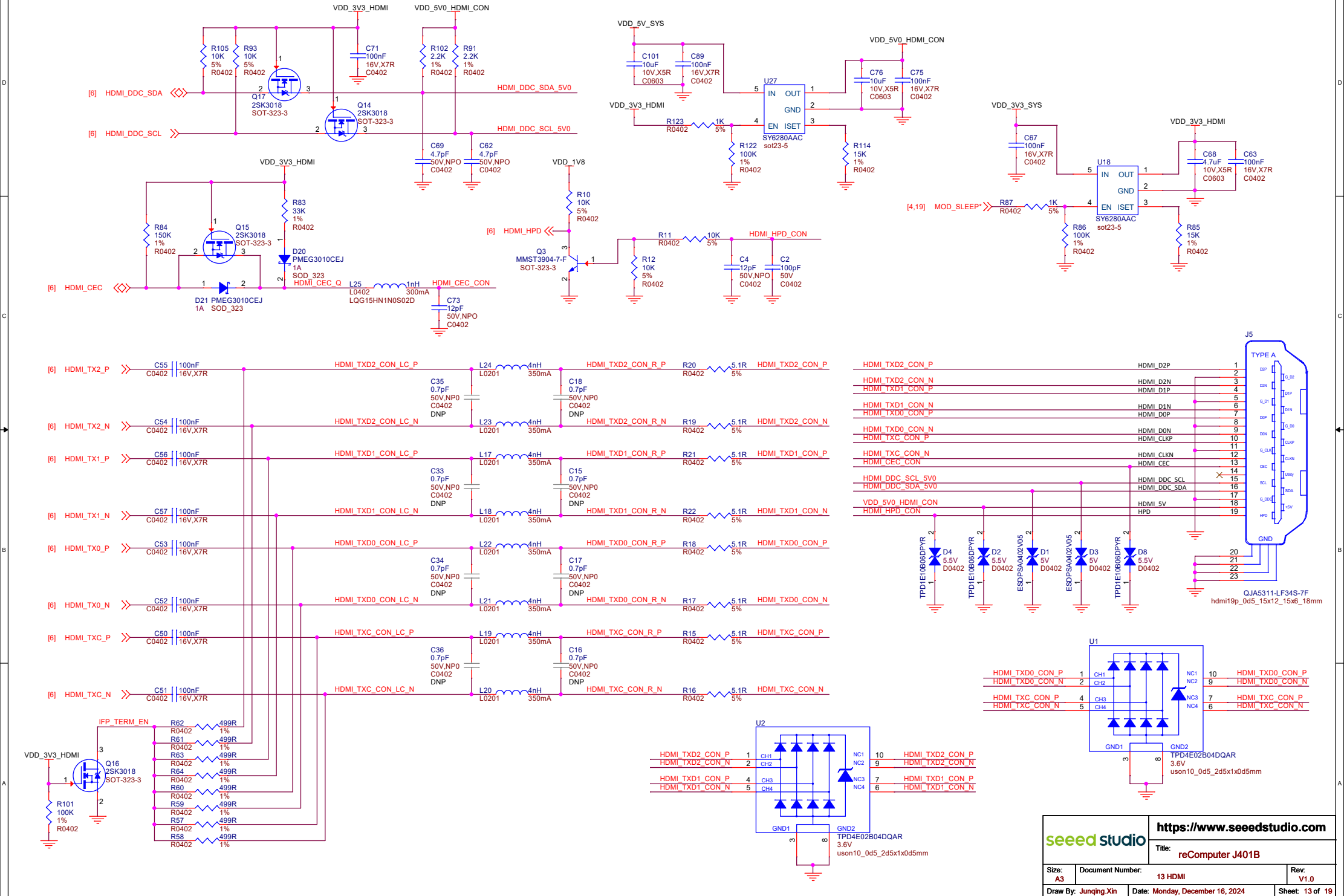
USB3.2 TYPE-A x2 (A)



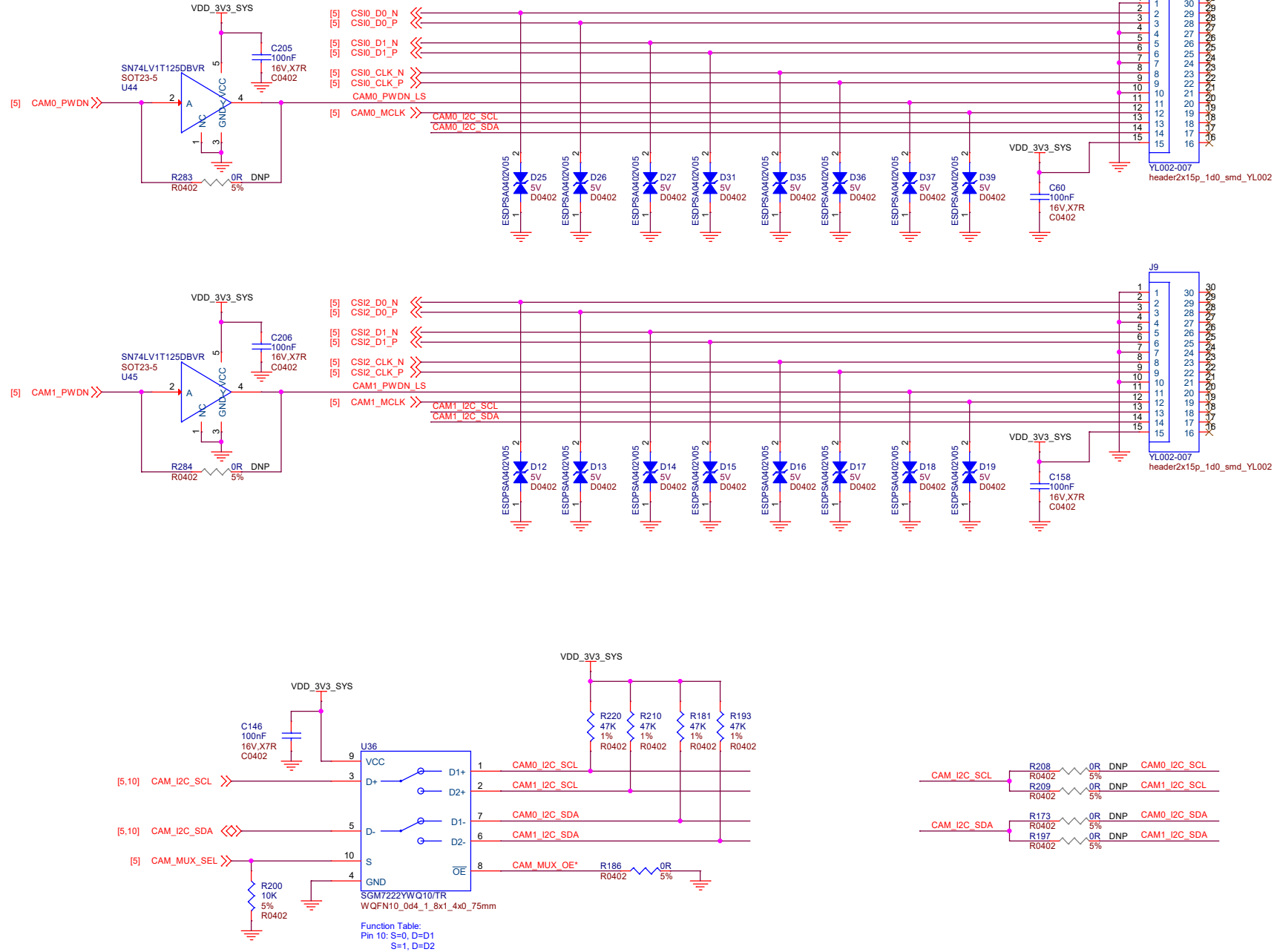
USB3.2 TYPE-A x2 (B)



HDMI Connector



CSI CAM CONNECTOR



The schematic diagram illustrates the internal components and connections of the Gigabit Ethernet module. The module is a J6 connector with 20 pins. The internal components include four RJ45 ports (J1, J2, J3, J4), four RJ45 ports (J5, J6, J7, J8), four RJ45 ports (J9, J10, J11, J12), and four RJ45 ports (J13, J14, J15, J16). The module is powered by a 5V supply (C9) and a 3V3 supply (C7). The module is connected to a PCB with various components including resistors (R1, R277, R402, R278, R279, R280, R281, R402), capacitors (C1, C0402, C7, C0402), and LEDs (GREEN_LED, YELLOW_LED). The module is also connected to a shield (SHIELD_1, SHIELD_2) and a ground (CGND).

USB Type C USB2.0 Only

J1
USB2.0 Type C
USB2_0 TYPE C long

A4/B9/VBUS
A9/B4/VBUS

CC1
CC2

DN1
DN2

DP1
DP2

S1
S2
S3
S4
S5
S6

A1/B12/GND
A12/B1/GND

B9
B4

A5
B5

A7
B7

A6
B6

A8
B8

B12
B1

R2
R3

R4
R0402

C3
100nF
16V X7R
C0402

D5
5.5V
D0402

C5
4.7uF
10V X5R
C0603
DNP

R7
10K
5%
R0402

TPD1E10B00CPYR

ESDPSA0402V05

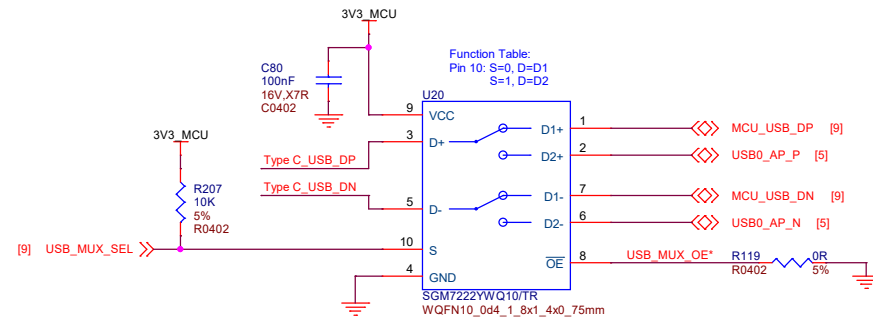
D6
5V
D0402

D7
5V
D0402

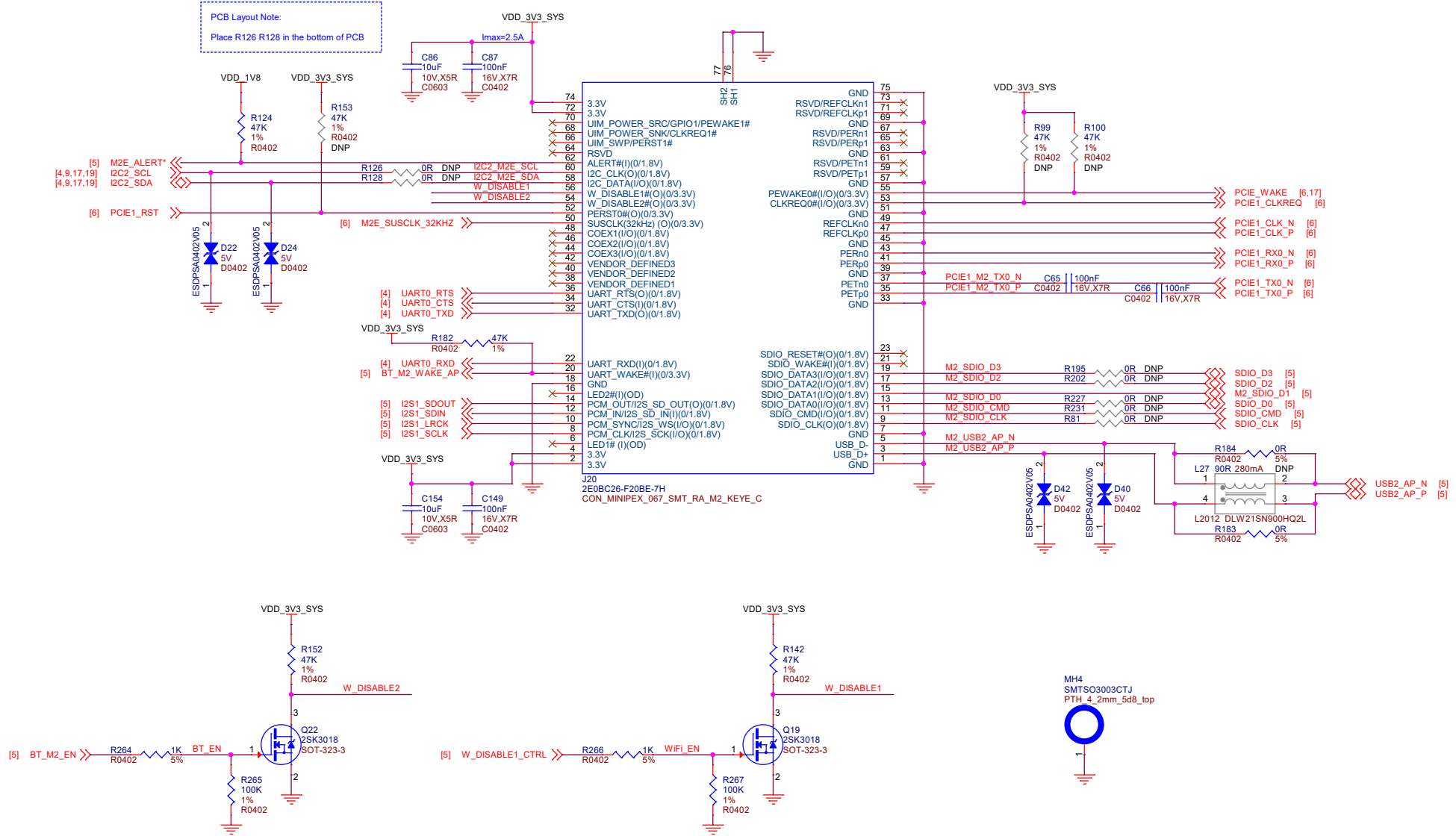
Q1
MMST3904-7-F
SOT-323-3

USB0_VBUS_DET

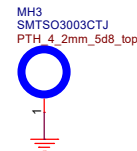
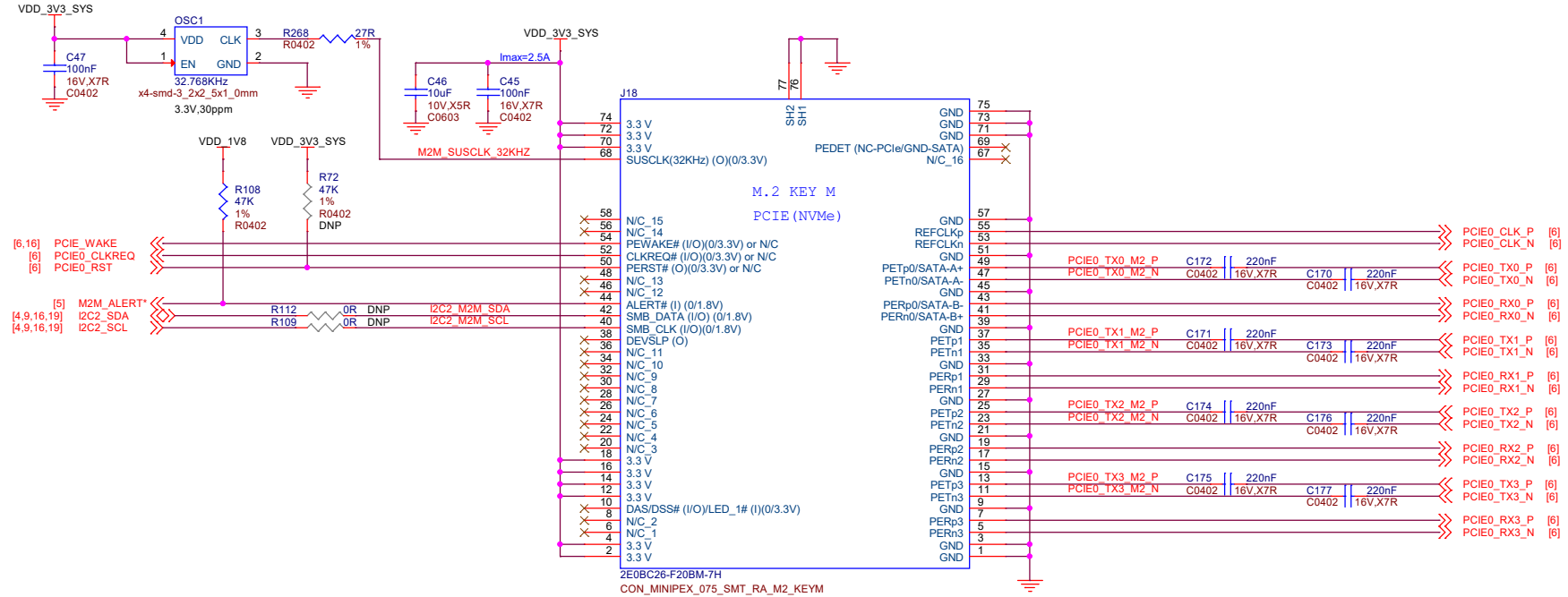
[9] USB_MUX_



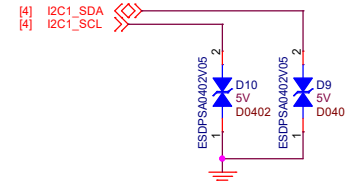
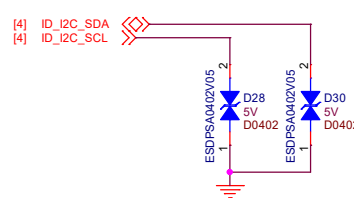
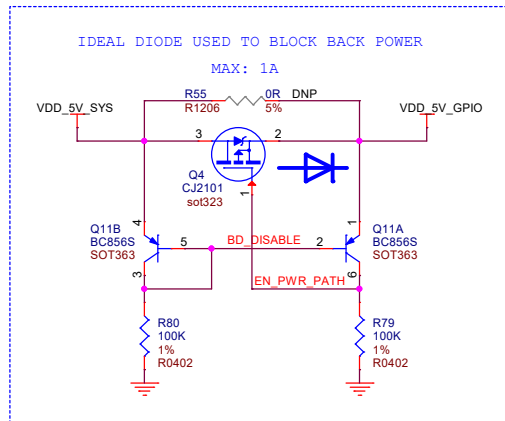
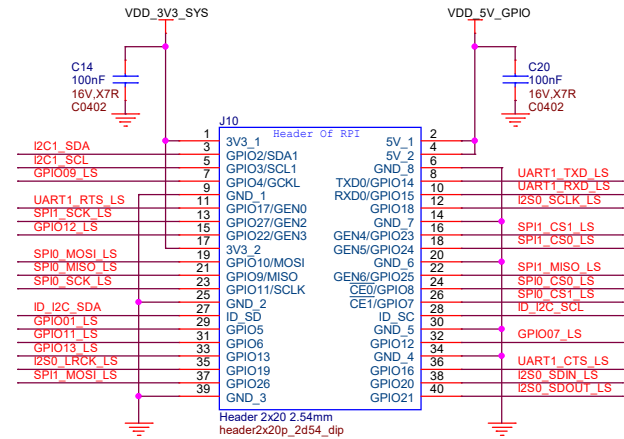
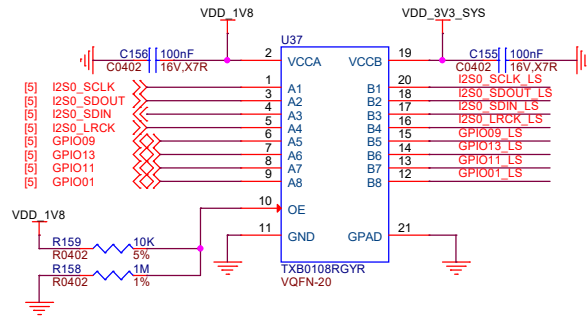
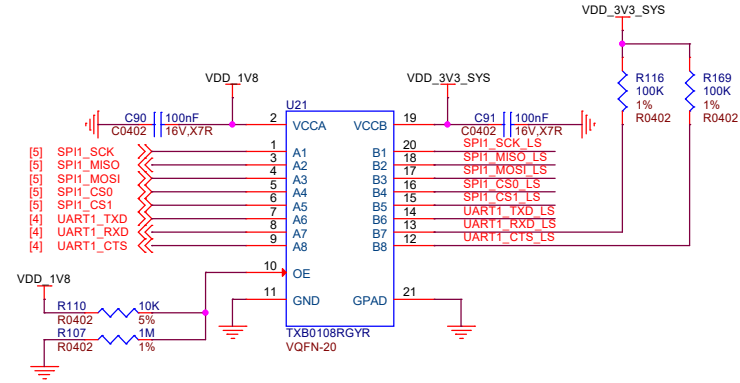
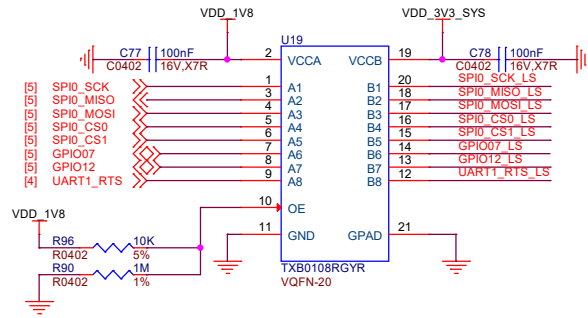
M.2 KEY-E (WIFI/BT)



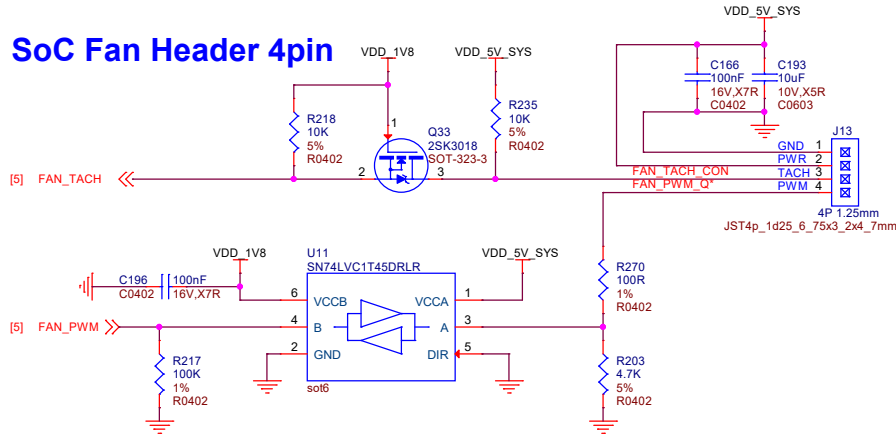
M.2 KEY-M (NVME)



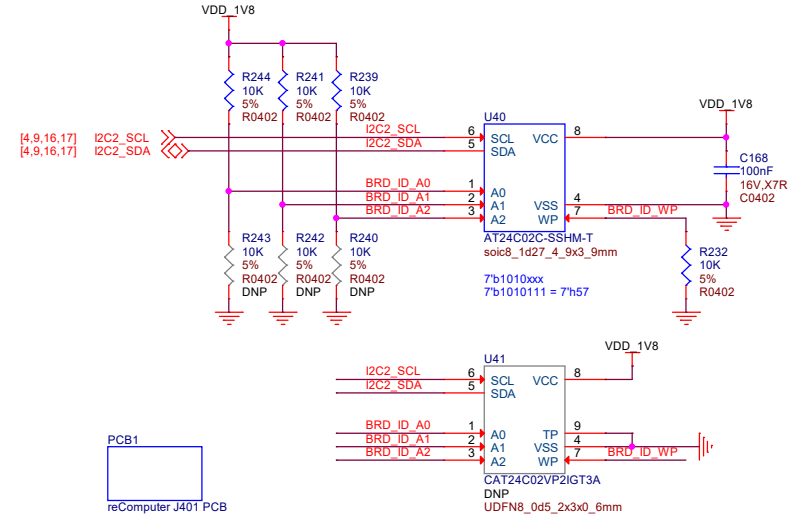
40PIN GPIO EXPANSION HEADER & LEVEL SHIFTERS



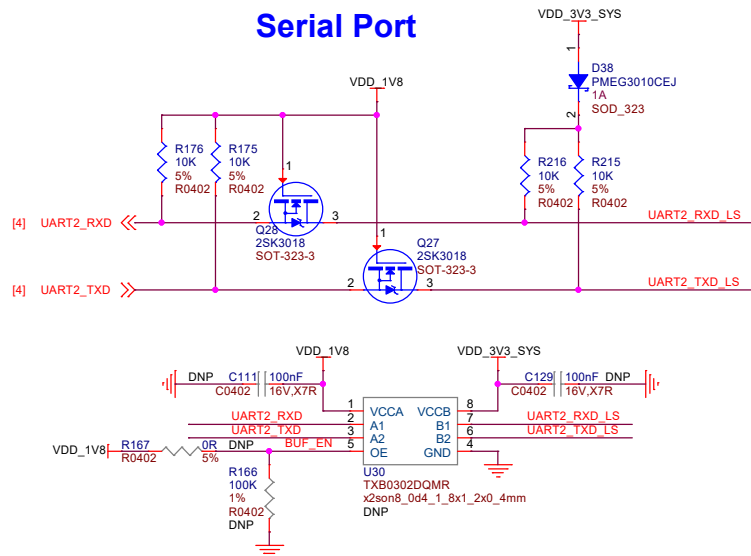
SoC Fan Header 4pin



Carrier Board Config



Serial Port



Optional Button Header Pinout

